



Zakonodajko: Retrieval-Augmented Question Answering over Slovenian Tax Legislation

ByteBanda project team

Abstract

Zakonodajko is a Retrieval-Augmented Generation (RAG) assistant for Slovenian tax legislation. It extracts local PISRS HTML documents, chunks them, embeds the chunks with a multilingual sentence-transformer, indexes them in FAISS, retrieves grounded context, and prompts a local instruction-tuned language model. We compare a fixed dense baseline with a Version 2 pipeline that adds article-aware legal chunking and hybrid reranking. The corrected aggregate results from the existing report show that Version 2 improves source hit@3 from 40.0% to 90.0% for the full 30-question run, while the strict 20-question prompt run reaches 95.0% source hit@3, 40.0% article hit@3, and 0.65/2 manual correctness.

Keywords

retrieval-augmented generation, Slovenian legislation, tax law, FAISS, sentence embeddings

Advisor: Slavko Žitnik

Introduction

Legal question answering requires exact source grounding: a wrong article, exception, threshold, or act can change the answer. Our project, *Zakonodajko*, addresses this by using RAG over Slovenian tax legislation instead of relying on model memory alone. The collection contains local PISRS versions of ZDDV-1, ZDoh-2, ZDDPO-2, ZDavP-2, and related implementing rulebooks. We kept the system retrieval-first because the main bottleneck was retrieval precision, not model capacity.

Methods

The baseline system splits documents into overlapping 1,200-character windows with 200 characters of overlap, embeds chunks with multilingual MiniLM, normalizes vectors, and searches a FAISS IndexFlatIP index with `top_k=3`. Version 2 keeps dense retrieval as the candidate generator but changes two parts of the pipeline. First, article-aware chunking detects headings such as `3. člen` and `113.a člen`, then stores source filename, law identifier, document role, article number, article title, and section position as metadata. Second, hybrid reranking scores the `top candidate_k=100` dense candidates as

$$s(q, c) = s_{\text{dense}}(q, c) + 0.25s_{\text{lexical}}(q, c) + s_{\text{law}}(q, c),$$

where the lexical term is normalized token overlap over chunk text plus metadata, and the law boost is 0.20 when the question names or strongly implies the same legal act as the chunk.

The generation step renders the retrieved chunks with source, chunk ID, score, law ID, article number, and title. We evaluated the baseline prompt, which asks the model to answer only from context and cite sources, and a strict prompt, which tells the model to prefer the chunk whose act and article directly answer the question and to say the answer is not found when the exact rule is missing. Retrieval is evaluated with source hit@3, article hit@3, phrase hit@3, and manual context relevance. Generation is scored manually from 0 to 2 for faithfulness and correctness.

Table 1. Legal documents used in the RAG evaluation.

ID	Document
ZDDV-1	Value Added Tax Act and rulebook
ZDoh-2	Personal Income Tax Act
ZDDPO-2	Corporate Income Tax Act and rulebook
ZDavP-2	Tax Procedure Act and rulebook

Results

Table 2 uses the aggregate values from Table 2 of the existing report.pdf. The corrected values show that the full Version 2 legal-hybrid retrieval run improves source hit@3 to

0.90, article hit@3 to 0.30, and phrase hit@3 to 0.35 over 30 questions. The strict prompt was available for 20 questions and reaches the best source and article scores in the table: 0.95 source hit@3 and 0.40 article hit@3.

Table 2. Automatic evaluation metrics from the existing report results. Src/Art/Phr are @3 metrics; Ctx/Fth/Cor use 0–2 scores.

Run	n	Src	Art	Phr	Ctx	Fth	Cor
V1 old	30	0.40	0.10	0.15	0.10	1.00	0.25
V2 fixed	30	0.45	0.15	0.20	0.40	1.05	0.30
V2 hybrid	30	0.90	0.30	0.35	1.30	1.15	0.40
V2 base	30	0.90	0.35	0.35	1.30	1.45	0.65
V2 strict	20	0.95	0.40	0.35	1.30	1.05	0.80

The most important improvement is source-level retrieval: the legal-hybrid retriever usually selects the correct act, while the fixed dense runs often retrieve semantically related material from rulebooks or another law. Article retrieval remains the main limitation. Even in the best strict-prompt run, article hit@3 is 0.40, so the system often finds the right law but not the exact definitional or procedural article.

Table 3. Manual generation assessment from Table 3 of the existing report. Scores use a 0–2 scale over 20 questions.

Metric	Baseline prompt	Strict prompt
Context relevance	1.20	1.25
Faithfulness	1.45	1.55
Correctness	0.60	0.65

Table 3 shows that strict prompting gives only a small correctness gain. This is expected because both prompt variants depend on the same retrieved context. When the correct article is present, both prompts can answer well; when retrieval returns only related material, the model either gives a faithful but legally wrong answer or refuses because the exact rule is missing. The common error patterns are: correct law but wrong article, rulebook provisions instead of the statute definition, cross-law confusion between ZDoh-2 and ZDDPO-2, and faithful answers to irrelevant retrieved passages.

Discussion and Conclusion

Version 2 is a clear improvement over the initial dense baseline because legal chunk metadata and hybrid reranking make source selection much stronger. The remaining weakness is exact article retrieval, not answer fluency. For legal RAG, topical similarity is insufficient; the system must retrieve the precise rule before generation starts. The next step should therefore be a real sparse component, such as BM25 over all chunks, combined with dense retrieval and stronger boosts for article titles and article numbers. The evaluation should also move away from old fixed-chunk IDs and use source, article, and phrase expectations as the stable metrics.

Overall, Zakonodajko demonstrates a reproducible RAG loop for Slovenian tax law, from local legal documents through chunking, FAISS indexing, retrieval, generation, and structured evaluation. The corrected results show strong source-

level progress, but article hit@3 and manual correctness are still too low for dependable legal question answering.